

Q1: 24 Feb (Shift 1) - Numerical

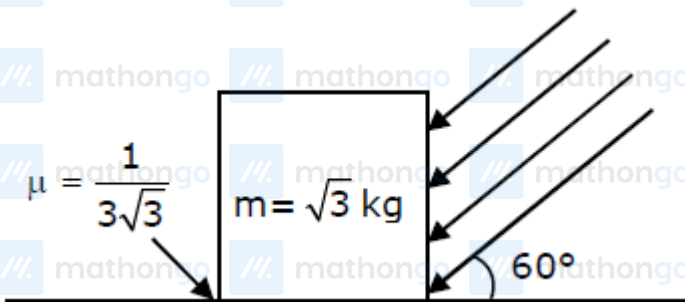
The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force that should be applied on the block to keep it adhere to the wall will be N
 $[g = 10 \text{ ms}^{-2}]$

Q2: 24 Feb (Shift 1) - Numerical

An inclined plane is bent in such a way that the vertical cross-section is given by $y = \frac{x^2}{4}$ where y is in vertical and x in horizontal direction. If the upper surface of this curved plane is rough with coefficient of friction $\mu = 0.5$, the maximum height in cm at which a stationary block will not slip downward is ____ cm.

Q3: 26 Feb (Shift 1) - Numerical

As shown in the figure, a block of mass $\sqrt{3}$ kg is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be $3x$. The value of x will be $\left[g = 10 \text{ m/s}^2; \sin 60^\circ = \frac{\sqrt{3}}{2}; \cos 60^\circ = \frac{1}{2} \right]$ (round off to nearest integer)



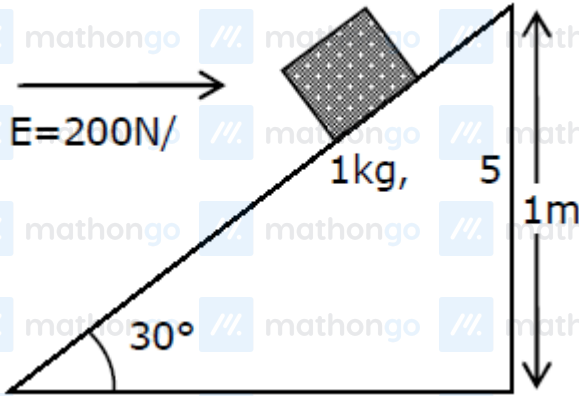
Q4: 26 Feb (Shift 2) - Single Correct

An inclined plane making an angle of 30° with horizontal is placed in a uniform horizontal electric field $200 \frac{\text{N}}{\text{C}}$ as shown in the figure. A body of mass 1 kg and charge 5 mC is allowed to slide down from rest at a height of 1 m. If the coefficient of friction is 0.2, find the time taken by the body to reach the bottom.

Questions with Answer Keys

MathonGo

$$\left[g = 9.8 \text{ m/s}^2, \sin 30^\circ = \frac{1}{2}; \cos 30^\circ = \frac{\sqrt{3}}{2} \right]$$



(1) 2.3 s

(2) 0.46 s

(3) 1.3 s

(4) 0.92 s

Answer Key

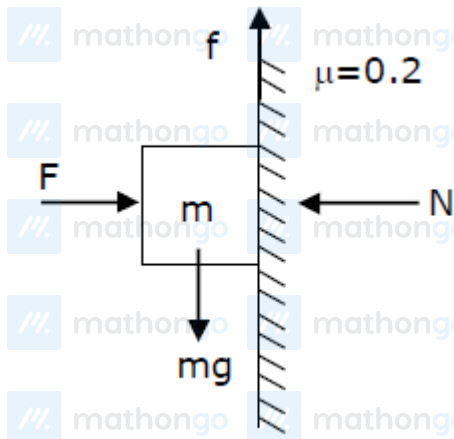
Q1 (25)

Q2 (25)

Q3 (3)

Q4 (3)

Q1 (25)



Given : $\mu_s = 0.2$

$m = 0.5 \text{ kg}$

$g = 10 \text{ m/s}^2$

we know that

$$f_s = \mu N \text{ and } \dots (1)$$

To keep the block adhere to the wall

here $N = F \dots (2)$

$$f_s = mg \dots (3)$$

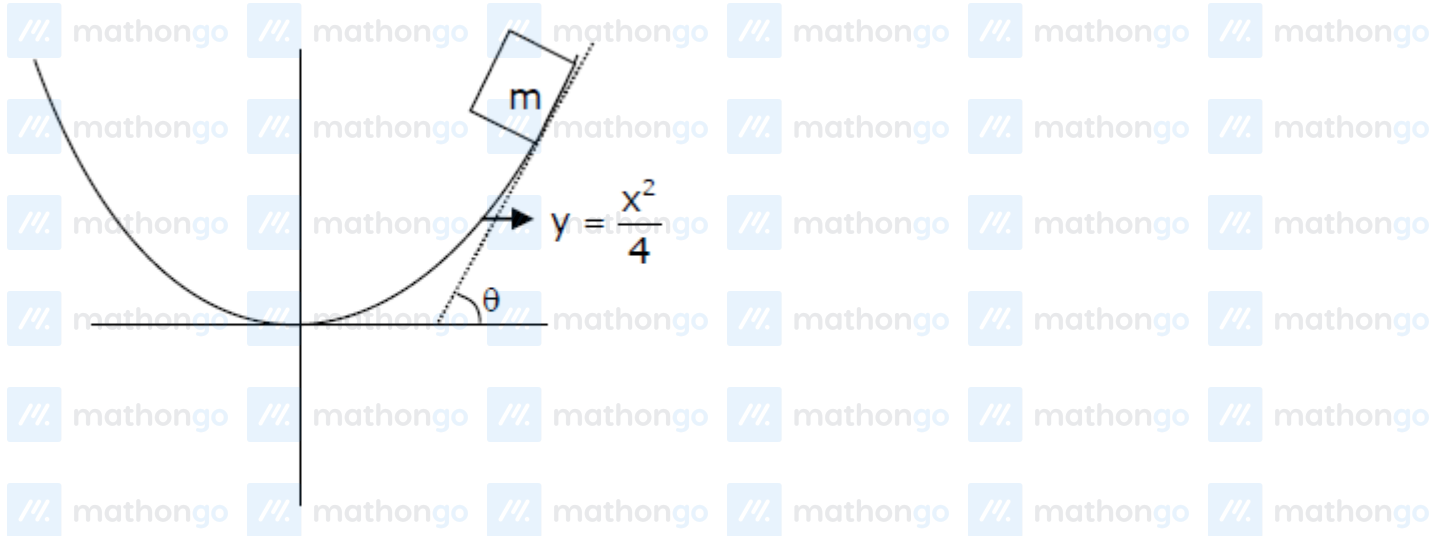
from equation (1), (2), and (3), we get

$$\Rightarrow mg = \mu F$$

$$\Rightarrow F = \frac{mg}{\mu} \Rightarrow F = \frac{0.5 \times 10}{0.2}$$

$$F = 25 \text{ N}$$

Q2 (25)



given

$$y = \frac{x^2}{4}$$

$$\mu = 0.5$$

condition for block will not slip downward

$$mg \sin \theta = \mu mg \cos \theta$$



$$\Rightarrow \tan \theta = \mu$$

and we know that

$$\Rightarrow \tan \theta = \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \mu \Rightarrow \frac{x}{2} = 0.5$$

$$\left[y = \frac{x^2}{4} \right]$$

$$\left[\frac{dy}{dx} = \frac{x}{2} \right]$$

$$\Rightarrow x = 1$$

put $x = 1$ in equation $y = x^2/4$

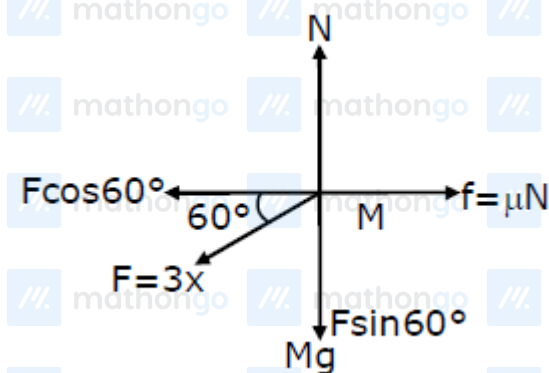
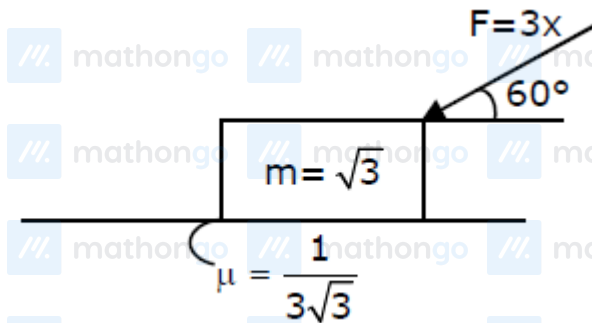
Hints and Solutions

MathonGo

$$\Rightarrow y = \frac{(1)^2}{4} \Rightarrow y = \frac{1}{4} \Rightarrow y = 0.25$$

$$y = 25 \text{ cm}$$

Q3 (3)



$$N = Mg + F \sin 60^\circ$$

$$N = \sqrt{3}g + \frac{F\sqrt{3}}{2}$$

For No slipping

$$F \cos 60^\circ = \text{Friction}$$

$$\frac{F}{2} = \mu N = \frac{1}{3\sqrt{3}} \left(\sqrt{3}g + \frac{F\sqrt{3}}{2} \right)$$

$$\frac{F}{2} = \frac{g}{3} + \frac{F}{6}$$

$$\frac{F}{2} - \frac{F}{6} = \frac{g}{3}$$

Hints and Solutions

MathonGo

$$\frac{6F-2F}{12} = \frac{g}{3}$$

$$4F = 4g$$

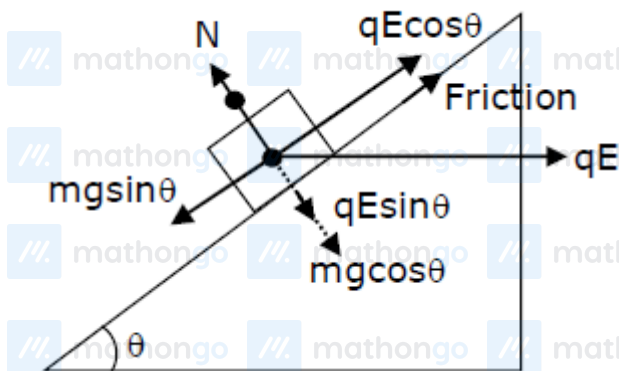
$$F = 10$$

$$F = 3x$$

$$X = \frac{F}{3} = \frac{10}{3} = 3.33$$

$$X = 3.33$$

Q4 (3)



$$F = mg \sin \theta - (\mu N + qE \cos \theta)$$

$$F = mg \sin \theta - \mu(mg \cos \theta + qE \sin \theta) - qE \cos \theta$$

$$F = 1 \times 10 \times \sin 30^\circ - 0.2(1 \times 10 \times \cos 30^\circ + 200 \times 5 \times 10^{-3} \sin 30^\circ)$$

$$- 200 \times 5 \times 10^{-3} \cos 30^\circ$$

$$F = 2.3 \text{ N}$$

$$a = \frac{F}{m} \Rightarrow \frac{2.3}{1} \Rightarrow 2.3 \text{ m/sec}^2$$

$$t = \sqrt{\frac{25}{9}} \Rightarrow \sqrt{\frac{2 \times 2}{2.3}} \Rightarrow 1.3 \text{ sec}$$